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APUL

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PHYSICS

1) Identify the pair of physical quantities, which have different dimensions :

(1) Wave number and Rydberg's constant
(2) Stress and coefficient of elasticity
(3) Coercivity and magnetisation
(4) Specific heat and latent heat

2) Density of a liquid in CGS system is 0.625 g/cm^3 its magnitude in SI system is :

(1) 0.625
(2) 0.0625
(3) 0.00625
(4) 625

3) Two numbers a = 0.92 and b = 0.08 are given. The number of significant figures present in the result after the following operations a + b and a – b :-

(1) 3, 3
(2) 3, 1
(3) 3, 2
(4) 2, 2

4)

Physical quantity $Q = \frac{x^2y^3}{z}$. If maximum precentage error in x, y and z are 1%, 2% and 3% respectively. Find % error in Q.

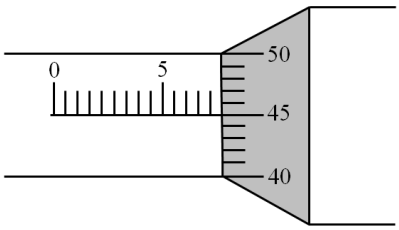
(1) 8%
(2) 7%
(3) 6%
(4) 11%

5) The vernier constant of a vernier calliper is 0.1 mm and it has zero error of (– 0.04) cm. While measuring the diameter of a sphere it was observed that main scale reading is 2.5 cm and 6th division of vernier coincides with any of main scale division. The corrected reading of diameter will be :

(1) 2.40 cm
(2) 2.42 cm
(3) 2.52 cm

(4) 2.60 cm

6) Reading of screw gauge will be if
→ Main scale has 1/2 mm marks.
→ Circular scale has 50 division.



→ In complete rotation C.S moves 2 mm.

(1) 5.4 mm
(2) 6.7 mm
(3) 8.8 mm
(4) 10.5 mm

7) Force F is given by $F = ax + bt^2$, where t is time and x is Displacement then Dimensions of a and b are -

(1) $M^1L^1T^{-3}$, $M^1L^1T^{-4}$
(2) $M^1L^0T^{-2}$, $M^1L^{-1}T^{-3}$
(3) $M^1L^0T^{-2}$, $M^1L^1T^{-4}$
(4) $M^0L^1T^{-2}$, $M^1L^0T^{-4}$

8) A student measured the length of a rod and wrote it as 3.60 cm. Which instrument did he use to measure it ?

(1) A screw gauge having 100 divisions in the circular scale and pitch as 1mm.
(2) A screw gauge having 50 divisions in the circular scale and pitch as 1mm
(3) A meter scale
(4) A vernier calliper where the 10 divisions in vernier scale match with a division in the main scale and main scale have 10 divisions in 1 cm.

9) If dimensions of x and y are different, then which of the following operation is valid :-

(1) x – y
(2) x + y
(3) x/y
(4) $e^{-x/y}$

10) The position-time relation of a particle is given by $x = (t + 2)^{-1}$. Then its acceleration is proportional to $(\text{velocity})^n$. Find the value of n :-

(1) $\frac{3}{2}$

(2) $-\frac{3}{2}$

(3) –3

(4) –2

11) A stone is dropped into a well in which the level of water is h below the top of the well. If 2v is velocity of sound, the time T after which the splash is heard is given by :

(1) $T = \frac{2h}{v}$

(2) $T = \sqrt{\frac{2h}{g}} + \frac{h}{2v}$

(3) $T = \sqrt{\frac{2h}{v}} + \frac{h}{v}$

(4) $T = \sqrt{\frac{h}{2g}} + \frac{2h}{v}$

12) Two balls A and B are thrown with speed u and u/2 respectively. Both the balls cover the same horizontal distance before returning to the plane of projection. If the angle of projection of ball B is 15° with the horizontal, then the angle of projection of A is :

(1) $\sin^{-1}\left(\frac{1}{8}\right)$

(2) $\frac{1}{2}\sin^{-1}\left(\frac{1}{8}\right)$

(3) $\frac{1}{3}\sin^{-1}\left(\frac{1}{8}\right)$

(4) $\frac{1}{4}\sin^{-1}\left(\frac{1}{8}\right)$

13) A body thrown upwards from ground, experience air resistance of 2 m/s^2. The ratio of time of ascent to time of descent is $\sqrt{\frac{\alpha}{\beta}}$; α, β are integers. Then minimum value of α + β is (g = 10 m/s^2)

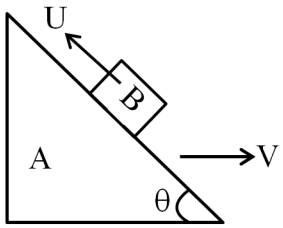
(1) 6

(2) 5

(3) 4

(4) 3

14) A block B moves with a velocity U relative to the wedge A. If the velocity of the wedge is V (as shown in figure) for the block to move vertically up relative to ground, θ is –



(1) $\cos^{-1}\left(\frac{U}{V}\right)$

(2) $\sin^{-1}\left(\frac{U}{V}\right)$

(3) $\tan^{-1}\left(\frac{U}{V}\right)$

(4) $\cos^{-1}\left(\frac{V}{U}\right)$

15) A particle is projected at an angle of 30° w.r.t. horizontal with speed 20 m/s then angle between velocity vector and position vector at t = 1 sec is -

(1) $\theta = \cos^{-1}\left(2\sqrt{\frac{3}{13}}\right)$

(2) $\theta = \cos^{-1}\left(2\sqrt{\frac{5}{13}}\right)$

(3) $\theta = \cos^{-1}\left(2\sqrt{\frac{17}{13}}\right)$

(4) $\theta = \cos^{-1}\left(2\sqrt{\frac{19}{13}}\right)$

16) **Statement-1 :** Projectile motion is an example of non-uniform motion.
Statement-2 : In projectile motion both magnitude and direction of velocity are variable.

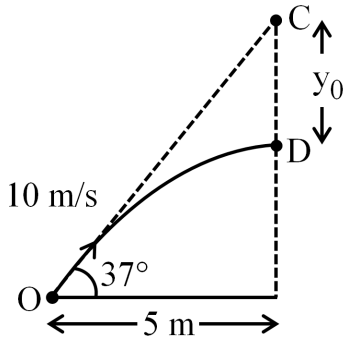
(1) Statement-1 is true, Statement-2 is true

(2) Statement-1 is false, Statement-2 is true

(3) Statement-1 is true, Statement-2 is false

(4) Statement-1 is false, Statement-2 is false

17) A ball is thrown from a point O aiming a target at angle 37° with the horizontal so that the ball may hit the target at C but the ball hits at point D, a vertical distance y_0 below C. If the initial velocity of the ball is 10 m/s and the horizontal distance between O and C is 5 m. Find y_0 :-



(1) 1 m

(2) 1.5 m

(3) 2.5 m

(4) 1.95 m

18) Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk down on a moving escalator in opposite direction.

(1) $\frac{t_1 t_2}{t_2 - t_1}$

(2) $\frac{t_1 t_2}{t_2 + t_1}$

(3) $t_1 - t_2$

(4) $\frac{t_1 + t_2}{2}$

19) Calculate the time an aeroplane having velocity 50 m/s takes to fly around a square with side 160 m. If the wind is blowing at velocity 30 m/s along one side of the square.

(1) 10sec

(2) 12sec

(3) 18sec

(4) 22sec

20) A car with a vertical wind shield moves in a rain strom at speed of 40 km/hr. The rain drops fall vertically with a terminal speed of 20 m/sec. The angle at which the rain drops strike the wind shield is-

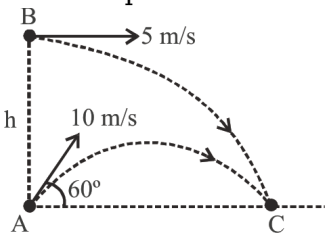
(1) $\tan^{-1}\left(\frac{5}{9}\right)$

(2) $\tan^{-1}\left(\frac{9}{5}\right)$

(3) $\tan^{-1}\left(\frac{3}{2}\right)$

(4) $\tan^{-1}\left(\frac{2}{3}\right)$

21) A particle A is projected from the ground with an initial velocity of 10 m/s at an angle of 60º with horizontal. From what height h should another particle B be projected horizontally with 5 m/s so that both the particles collide on ground at point C if both are projected simultaneously :



(1) 10 m

(2) 30 m

(3) 15 m

(4) 25 m

22) If a person (A) is moving with velocity $20\hat{i}+4\hat{j}$ and person (B) is moving with velocity $10\hat{i}+3\hat{j}$ calculate the velocity of (B) seen from (A) :-

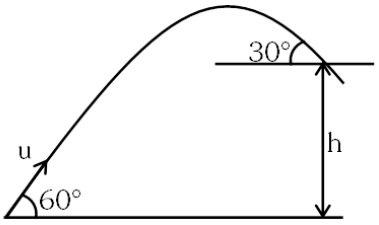
(1) $-10\hat{i}+4\hat{j}$

(2) $10\hat{i}-\hat{j}$

(3) $-10\hat{i}-\hat{j}$

(4) $10\hat{i}+\hat{j}$

23) A stone is projected at an angle of 60° from the ground level strikes at an angle of 30° on the roof of building of height h. Then speed of projection is :



(1) $\sqrt{2gh}$

(2) $\sqrt{6gh}$

(3) $\sqrt{3gh}$

(4) $\sqrt{gh}$

24) A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30° with the horizontal. The height of the tower is 70 m. After how much time from the instant of throwing will the ball reach the ground ?

(1) 2 s

(2) 5 s

(3) 7 s

(4) 9 s

25) If two balls are projected with speeds 90m/s and 40m/s from top of a tower simultaneously in opposite directions both balls are projected horizontally then find time after which their velocities become perpendicular to each other.

(1) 2 sec.

(2) 4 sec.

(3) 9 sec.

(4) 6 sec.

26) If a car covers $\frac{2}{5}^{th}$ of the total distance with speed v_1 and $\frac{3}{5}^{th}$ distance with speed v_2, then the average speed is–

(1) $\frac{1}{2}\sqrt{v_1 v_2}$

(2) $\frac{v_1 + v_2}{2}$

(3) $\frac{2v_1 v_2}{v_1 + v_2}$

(4) $\frac{5v_1 v_2}{3v_1 + 2v_2}$

27) A projectile is given an initial velocity of $\hat{i} + 2\hat{j}$ .The cartesian equation of its path is (g = 10m/s^2).

(1) $y = 2x - 5x^2$

(2) $y = x - 5x^2$

(3) $4y = 2x - 5x^2$

(4) $y = 2x - 25x^2$

28) A particle is projected from level ground. Assuming projection point as origin, x-axis along horizontal and y-axis along vertically upwards. If particle moves in x-y plane and its path is given by $y = \frac{3x}{4}\left(1 - \frac{x}{240}\right)$. Then match the physical quantities given in column-I with the values given in column-II.

	Column-I		**Column-II**
(A)	Horizontal component of velocity	(p)	240
(B)	Time of flight	(q)	45
(C)	Maximum height	(r)	40
(D)	Horizontal range	(s)	6

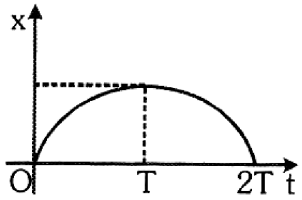
(1) A–r; B–s; C–q; D–p

(2) A–r; B–s; C–p; D–q

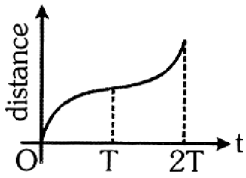
(3) A–s; B–r; C–q; D–p

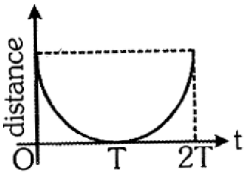
(4) A–s; B–r; C–p; D–q

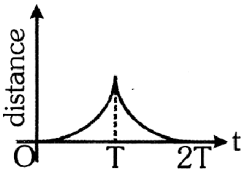
29) The x – t graph of a particle moving along a straight line is shown in figure.

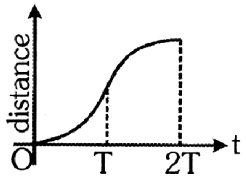


The distance-time graph of the particle is correctly shown by :

(1) 

(2) 

(3) 

(4) 

30) Two bodies are thrown vertically upwards with their initial speeds in the ratio 2 : 3. The ratio of the maximum heights reached by them and the ratio of time taken by them to return back to the ground respectively are -

(1) 4 : 9 and 2 : 3

(2) 2 : 3 and $\sqrt{2} : \sqrt{3}$

(3) $\sqrt{2} : \sqrt{3}$ and 4 : 9

(4) None of these

31) A particle travels such that $\frac{dV}{dt} = -a\sqrt{v}$ where a is constant. Its initial velocity is v^0. The time at which it stops is :-

(1) $\frac{2\sqrt{V_0}}{a}$

(2) $\frac{\sqrt{V_0}}{a}$

(3) $2a\sqrt{V_0}$

(4) $a\sqrt{V_0}$

32) A particle leaves the origin with initial velocity $\vec{v}_0 = 11\hat{i} + 14\hat{j}$ m/s. It undergoes a constant acceleration given by $\vec{a} = -\frac{22}{5}\hat{i} + \frac{2}{15}\hat{j}$ m/s^2. At the instant when particle crosses y-axis, direction in which particle is moving is :-

(1) At angle 37° from +x-axis towards +y-axis
(2) At angle 37° from –x-axis towards +y-axis
(3) At angle 53° from +x-axis towards +y-axis
(4) At angle 53° from –x-axis towards +y-axis

33) Which of following statement is false :-

(1) Displacement is zero but particle may have some distance
(2) Distance is zero, but displacement is non zero
(3) Speed is fixed but velocity can be variable for a body
(4) None of these

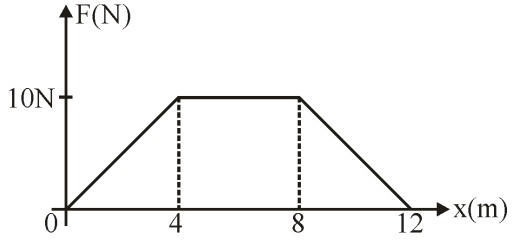
34) A spring of force constant k is cut in two parts at its one third length. When both the parts are stretched by same amount, the work done in the two parts, will be

(1) equal in both
(2) greater for the longer part
(3) greater for the shorter part
(4) data insufficient

35) The momentum of a body is increased by 25%. The kinetic energy is increased by about :

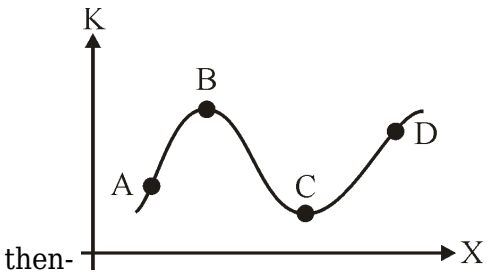
(1) 25%
(2) 5%
(3) 56%
(4) 38%

36) A particle of mass 0.1 Kg is subjected to a force which varies with distance as shown. If it starts its journey from rest at x = 0, then its velocity at x = 12 m is :-



(1) 0
(2) $20\sqrt{2}$ m/s
(3) $20\sqrt{3}$ m/s
(4) 40 m/s

37) For a particle moving under the action of a variable force, kinetic energy, position graph is given



then-

(1) At A particle is decelerating

(2) At B particle is accelerating

(3) At C particle has max. velocity

(4) At D particle has max. acc.

38) The only force acting on a block is along x-axis is given by F = $-\left(\frac{4}{x^2+2}\right)$N. when the block moves from x = –2m to x = 4m, the change in kinetic energy of block is-

(1) Positive

(2) Negative

(3) Zero

(4) May be +ve or – ive

39) An open knife edge of mass 'm' is dropped from a height 'h' on a wooden floor. If the blade penetrates upto depth 'd' into the wood, the average resistance offered by the wood to the knife edge is

(1) mg

(2) $mg\left(1-\frac{h}{d}\right)$

(3) $mg\left(1+\frac{h}{d}\right)$

(4) $mg\left(1+\frac{h}{d}\right)^2$

40) A bullet moving with a speed of 100 ms^{-1} can just penetrate two planks of equal thickness. Then the number of such planks penetrated by the same bullet when the speed is doubled will be :

(1) 4

(2) 8

(3) 6

(4) 10

41) Choose the correct statement among the following about the conservative force.
(a) Work is independent on path
(b) Work done in a closed loop is zero
(c) Work done against conservative force is stored in the form of potential energy

(1) Only a and b are correct

(2) Only b and c are correct

(3) Only c and a are correct

(4) a, b and c are correct

42) A particle in a certain conservative force field has a potential energy given by $U = \frac{20xy}{z}$. The force exerted on it is :-

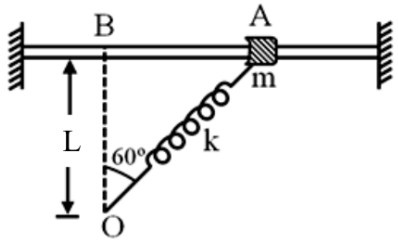
(1) $\left(\frac{20y}{z}\right)\hat{i} + \left(\frac{20x}{z}\right)\hat{j} + \left(\frac{20xy}{z^2}\right)\hat{k}$

(2) $-\left(\frac{20y}{z}\right)\hat{i} - \left(\frac{20x}{z}\right)\hat{j} + \left(\frac{20xy}{z^2}\right)\hat{k}$

(3) $-\left(\frac{20y}{z}\right)\hat{i} - \left(\frac{20x}{z}\right)\hat{j} - \left(\frac{20xy}{z^2}\right)\hat{k}$

(4) $\left(\frac{20y}{z}\right)\hat{i} + \left(\frac{20x}{z}\right)\hat{j} - \left(\frac{20xy}{z^2}\right)\hat{k}$

43) Find out speed of the ring at point B when released from point A in given figure. Natural length



of spring is L.

(1) $\frac{L}{4}\sqrt{\frac{k}{m}}$

(2) $\frac{L}{2}\sqrt{\frac{k}{m}}$

(3) $\frac{L}{3}\sqrt{\frac{k}{m}}$

(4) $L\sqrt{\frac{k}{m}}$

44) Force acting on a particle varies with distance travelled by a the particle as $F \propto \frac{1}{x^{1/3}}$, then power due to this force varies with x as:-

(1) $X^{1/3}$

(2) $X^{1/2}$

(3) $X^{-1/3}$

(4) X^{0}

45) Two man with weights in ratio 4:3 run up a staircase in time with ratio 12:11. The ratio of the power of first to that of second is :

(1) 4/3

(2) 12/11

(3) 48/33

(4) 11/9

CHEMISTRY

1) The coordination geometry around the manganese in decacarbonyldimanganese (0) is

(1) Trigonal bipyramidal

(2) Square pyramidal

(3) Square planar

(4) Octahedral

2) Match the columns.

	Column I (Coordination compound)		**Column II (Central metal atom)**
(A)	Chlorophyll	(p)	Rhodium
(B)	Blood pigment	(q)	Cobalt
(C)	Wilkinson catalyst	(r)	Calcium
(D)	Vitamins B_{12}	(s)	Iron
		(t)	Magnesium

(1) A-(t), B-(s), C-(p), D-(q)

(2) A-(s), B-(q), C-(p), D-(r)

(3) A-(p), B-(q), C-(r), D-(s)

(4) A-(r), B-(t), C-(p), D-(q)

3) Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?

(1) $[Co(NH_3)_3Cl_3]$

(2) $[Co(NH_3)_4Cl_2]$ Cl

(3) $[Co(NH_3)_6]Cl_3$

(4) $[Co(NH_3)_5Cl]$ Cl

4)

The correct order of the wavelength of light absorbed by the following complexes is,

A. $[Co(NH_3)_6]^{3+}$ B. $[Co(CN)_6]^{3-}$

C. $[Cu(H_2O)_4]^{2+}$ D. $[Ti(H_2O)_6]^{3+}$

Choose the **correct** answer from the options given below:

(1) B < D < A < C

(2) B < A < D < C

(3) C < D < A < B

(4) C < A < D < B

5)

Which of the following are paramagnetic?

A. $[NiCl_4]^{2-}$ B. $Ni(CO)_4$

C. $[Ni(CN)_4]^{2-}$ D. $[Ni(H_2O)_6]^{2+}$

E. $Ni(PPh_3)_4$

Choose the **correct** answer from the options given below :

(1) A and C only

(2) B and E only

(3) A and D only

(4) A, D and E only

6) Match the following and choose correct option.

	Ligand		**Property**
(A)	ethylenediamine	(P)	bidentate ligand
(B)	hydrazene	(Q)	tridentate ligand
(C)	dimethylgyloximato(dmg^-)	(R)	Monodentate ligand
(D)	dien	(S)	can form hydrogen bond

	A	B	C	D
(1)	P	Q	R	S
(2)	P	R	Q	S
(3)	P	R	S	Q
(4)	P	S	Q	R

(1) 1

(2) 2

(3) 3

(4) 4

7) Match the column correctly ?

	Column-I (Coordination entity)		**Column-II (Color of light absorbed)**
(a)	$[Co(NH_3)_5C\square]^{2+}$	(P)	Ultraviolet
(b)	$[Co(CN)_6]^{3-}$	(Q)	Red
(c)	$[Cu(H_2O)_4]^{2+}$	(R)	Yellow
(d)	$[Co(NH_3)_5H_2O]^{+3}$	(S)	Blue, Green

(1) a→S, b→Q, c→R, d→P

(2) a→R, b→P, c→Q, d→S

(3) a→P, b→R, c→Q, d→S

(4) a→Q, b→S, c→R, d→P

8) Which one is incorrect.

(1) EAN of Fe in brown ring complex is 37

(2) $[Mn(CO)_6]$ can act as Reducing agent.

(3) Sum of oxidation state and number of ring in $[Cu(dien)_2]SO_4$ complex is 4

(4) Tetrahedral crystal field splitting is only 4/9 of octahedral complex.

9) Crystal field stabilization energy for high spin d^4 octahedral complex is :-

(1) $-0.6\ \Delta_0 + P$

(2) $-1.8\ \Delta_0$

(3) $-1.6\ \Delta_0 + P$

(4) $-0.6\ \Delta_0$

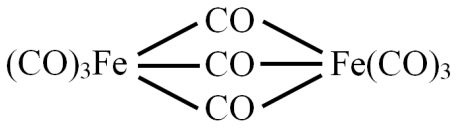
10) Which statement is not true :

(1) Red colour of $[Ni(dmg)_2]$ is due to d-d transition.

(2) $[Co(NH_3)_3Cl_3]$ complex shows fac-Mer isomerism

(3) $[Co(C_2O_4)_3]Cl_3$ complex is diamagnetic in nature

(4) Primary valency of complex $[Co(NH_3)_4Cl_2]Cl$ is one

11) Correct name of  is :

(1) Tri-μ-carbonyl bis[Tricarbonyliron(O)]

(2) Hexacarboxyliron(III)μ-tricarbonylferrate(O)

(3) Nonacarboxyl diiron(O)

(4) Tri-μ-carboxyl hexacarbonyliron(O)

12) Which ligand is used for remaining excess of copper and iron ?

(1) EDTA and DHG

(2) Cyanoamine and natphol

(3) D-pencillamine and desferrioxime B

(4) Ph_3P and NH_3

13) (A) $[Ni(H_2O)_5NH_3]^{+2}$
(B) $[Ni(H_2O)_4(NH_3)_2)]^{+2}$
(C) $[Ni(H_2O)_3(NH_3)_3)]^{+2}$

Correct statement

(1) (A) & (B) show's geometrical isomerism
(2) (A) & (C) show's geometrical isomerism
(3) (B) & (C) show's geometrical isomerism
(4) All show's geometrical isomerism

14) In the synergic bond of metal carbonyls :-

(1) π* molecular orbital of ligand is used
(2) π* molecular orbital of metal is used
(3) σ molecular orbital of ligand as well as metals are used
(4) σ* molecular orbital of metal is used

15) Match the column and choose the correct option.

	Column-I Complex Ion		**Column-II Wave length of absorbed light (NM)**
I.	$[CoCl(NH_3)_5]^{+2}$	A.	310
II.	$[Co(NH_3)_5(H_2O)]^{+3}$	B.	475
III.	$[Co(NH_3)_6]^{+3}$	C.	500
IV.	$[Co(CN_6)]^{-3}$	D.	535

(1) I – D; II – C; III – B; IV – A
(2) I – D; II – B; III – C; IV – A
(3) I – D; II – C; III – B; IV – A
(4) I – A; II – B; III – C; IV – D

16) Which of the following species show only stereoisomerism :

(1) $[Co(NH_3)_5Cl]Br$
(2) $[Pt(NH_3)_2Cl_2]$
(3) $[Co(NH_3)_4Cl_2]$
(4) Both (2) and (3)

17) **Assertion:–** Zeise's salt is a π-bonded organometallic compound.
Reason :- The oxidation number of plantinum in Zeise's salt is +2.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

18) Metal ion M^{2+} forms complexes with NH_3, CN^-, H_2O & en. Values of $K_{stability}$ of the complexes formed are 10^{12}, 10^{30}, 10^{16} & 10^{19} respectively. Hence

(1) CN^- is the strongest field ligand

(2) en is the strongest field ligand

(3) Strength of ligands cannot be predicted

(4) All ligands are equally strong.

19) Which is wrong statement -

(1) $[Co(NH_3)_6]$ $[Cr(CN)_6]$, $[Pt(NH_3)_4]$ $[CuCl_4]$ shows co-ordination isomerism

(2) Correct relation will be - $\Delta_0 > \Delta_{sp} > \Delta_t$

(3) In octahedral low spin complex having 5 electrons in d subshell the number of unpaired electrons is 1

(4) $[PtCl_4]^{-2}$ have dsp^2 hybridisation

20) Which of the following system has maximum number of unpaired electrons ?

(1) d^4 (octahedral, low spin)

(2) d^6 (tetrahedral)

(3) d^6 (octahedral, low spin)

(4) d^9 (octahedral)

21) No. of co-ordiation number, oxidation no., d- electron & unpaired d electron of metal in complex $[Co(NH_3)_5\ CO_3]\ ClO_4$ are respectively -

(1) 6 , 3 , 6 , 0

(2) 7 , 2 , 7 , 1

(3) 7 , 1 , 6 , 4

(4) 6 , 2 , 7 , 3

22) Correct statements regarding CFT are :
(a) The pattern of splitting depends upon the nature of the crystal field.
(b) During splitting degeneracy of d-orbitals is removed due to ligand electron metal electron repulsion.
(c) After splitting t_{2g}/t_2 always have lower while e_g/e have high energy.
(d) Crystal field splitting & Δ_0 depends upon the field produced by the ligand and charge on the metal ion.

(1) a, b, c

(2) b, c, d

(3) a, b, d

(4) a,b,c & d

23) Statements which are limitations of VBT of co-ordination compound.
(a) It does not give quantitative interpretation of magnetic data.
(b) Does not explain colour

(c) No quantitative interpretation of thermo-dynamic and kinetic stability.
(d) No exact prediction for tetrahedral and square planar structure
(e) Does not distinguish between weak and strong ligand.

(1) a, b & c

(2) a, b, c and d

(3) a, b, c, d and e

(4) only a and b

24) In absence of ligand, the substance (complex) will be colorless due to :

(1) No crystal field splitting

(2) Dehydration of complex

(3) Displacement of orbitals

(4) Inert nature of metal

25) Which of the following electronic arrangement is/are possible for inner orbital complex?
(I) $t_{2g}^3 e_g^2$ (II) $t_{2g}^6 e_g^1$ (III) $t_{2g}^3 e_g^0$ (IV) $t_{2g}^4 e_g^2$
Select the correct code:

(1) I, IV

(2) II, III

(3) III only

(4) III, IV

26) Which of the following complex can show ionisation, linkage, geometrical and optical, isomers?

(1) $[Co(NO_2)_2(en)_2]Br$

(2) $[Cr(NH_3)_4(CN)_2]Cl$

(3) $[Co(en)_2Cl_2]Br$

(4) $[Pt(NH_3)_3Cl_3]NO_2$

27) The correct order of ligands arranged in increasing field strength.

(1) $F^- < Br^- < I^- < NH_3$

(2) $H_2O < OH^- < CN^- < NH_3$

(3) $Br^- < F^- < H_2O < NH_3$

(4) $Cl^- < OH^- < Br^- < CN^-$

28) If an iron (III) complex with the formula $[Fe(NH_3)_x(CN)_y]^-$ has no electron in its e_g orbital, then the value of x + y is

(1) 6

(2) 4

(3) 3

(4) 5

29) In sodium nitroprusside the oxidation number, coordination number and effective atomic number of iron are respectively:

(1) +3, 6, 35
(2) +3, 6, 36
(3) +2, 3, 36
(4) +2, 6, 36

30) Neutral hexacarbonyl complex of which metal is most stable :

(1) Cr
(2) Mn
(3) V
(4) Ti

31)

Which is incorrect for primary valency & secondary valency, as mentioned infront of the compound ?

(S = Secondary Valency, P = Primary Valency)
(1) $[Cr(NH_3)_6]Cl_3$, P = 3, S = 6
(2) $K_2[PtCl_4]$, P = 2, S = 4
(3) $[Pt(NH_3)_2Cl_2]$, P = 2, S = 4
(4) $[Cu(NH_3)_4]SO_4$, P = 4, S = 4

32) Correct sequence of CO bond order in given compounds is :- (P) $Fe(CO)_5$ (Q) CO
(R) $H_3B \leftarrow CO$ (S) $[Mn(CO)_5]^-$

(1) P > R > S > Q
(2) S > P > R > Q
(3) Q > S > P > R
(4) R > Q > P > S

33) The IUPAC name of $[PtCl(NH_2–CH_3)(NH_3)_2]Cl$ is :-

(1) Diaminechloro(methylamine)platinum(II)chloride.
(2) (Dimethylamine)chlorodiamminoplatinum(II)chloride.
(3) Bis(ammine)chloro(methylamine)platinate(II)chloride.
(4) Diamminechloro(methylamine)platinum(II)chloride.

34) Which of the following will have three stereoisomeric forms :-
(i) $[Cr(NO_3)_3(NH_3)_3]$ (ii) $K_3[Co(C_2O_4)_3]$ (iii) $K_3[CoCl_2(C_2O_4)_2]$ (iv) $[CoBrCl(en)_2]$

(1) iii, iv
(2) i, ii and iv

(3) Only iv

(4) All

35) A complex $[MF_6]^{-x}$ has magnetic moment of 4.91 B.M. whereas $[M(CN)_6]^{-x}$ has zero magnetic moment. Metal ion can be :-

(1) Co^{+2}

(2) Mn^{+2}

(3) Fe^{+2}

(4) Fe^{+3}

36) Low spin tetrahedral complexes are not formed because

(1) for tetraheral complexes, the splitting energy is lower than pairing energy.

(2) for tetraheral complexes, the splitting energy is higher than pairing energy.

(3) electrons do not go to e_g in case of tetrahedral complexes

(4) tetrahedral complexes are formed by weak field ligands only.

37) Inner orbital complex which is paramagnetic in nature ?

(1) $[Fe(CO)_5]$

(2) $[Fe(CN)_6]^{-4}$

(3) $[Fe(H_2O)_5(NO)]SO_4$

(4) $[Fe(CN)_6]^{-3}$

38) Dimethyl glyoxime gives a red precipitate with Ni^{2+} which is used for its detection. To get this precipitate readily, the best pH range is :-

(1) < 1

(2) 2-3

(3) 3-4

(4) 9-11

39)

Consider the following reactions and identify the correct statement :-

$$[M(gly)_2(NH_3)_2]^+ \xrightarrow[-2NH_3]{+gly} [M(gly)_3]$$

(1) The hybridization of metal M in the product is d^2sp^3

(2) Stability increases

(3) Number of rings increases

(4) All of these

40) Choose incorrect stability order :

(1) $[Cu(NH_3)_4]^{2+} < [Cu(en)_2]^{2+} < [Cu(trien)]^{2+}$

(2) $[Fe(H_2O)_6]^{3+} < [Fe(NO_2)_6]^{3-} < [Fe(NH_3)_6]^{3+}$

(3) $[Co(H_2O)_6]^{3+} < [Rh(H_2O)_6]^{3+} < [Ir(H_2O)_6]^{3+}$

(4) $[Cr(NH_3)_6]^{1+} < [Cr(NH_3)_6]^{2+} < [Cr(NH_3)_6]^{3+}$

41) Some salts although containing two different metallic elements give test for one of them in solution, such salts are :-

(1) Complex salt

(2) Double salt

(3) Normal salt

(4) None of these

42) During chelation, which of following ligand form six membered ring :-

(1) glycinato

(2) oxalato

(3) acetyl acetonato

(4) bipyridyl

43) Pair of compound which is planar :-

(1) $[Ni(CN)_4]^{-4}$, $[PtCl_4]^{-2}$

(2) $[NiCl_4]^{-4}$, $[Ni(CN)_4]^{-2}$

(3) $[PtCl_4]^{-2}$, $[Ni(CO)_4]$

(4) $[Ni(CN)_4]^{-2}$, $[Ni(NH_3)_4]^{+2}$

44) Longest C–O Bond length is :

(1) $[Ni(CO)_4]$

(2) $[Co(CO)_4]^{-}$

(3) $[Fe(CO)_4]^{-2}$

(4) $[Mn(CO)_6]^{+}$

45) Which of the following is not organometallic compound :-

(1) RMgCl

(2) Ferocene

(3) $[Ni(CO)_4]$

(4) NaCN

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	3	4	4	3	3	4	3	1	2	2	2	4	1	1	4	1	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	3	3	4	4	1	1	1	1	1	4	2	3	3	4	4	4	3	2
Q.	41	42	43	44	45															
A.	4	2	4	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	1	2	3	3	2	3	4	4	1	3	3	1	1	4	2	1	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	3	1	2	1	3	1	4	1	4	4	4	1	3	1	4	4	4	2
Q.	86	87	88	89	90															
A.	1	3	4	3	4															

SOLUTIONS

PHYSICS

1) $S = \frac{Q}{m\Delta T} = \frac{J}{Kg^oC}$

$L = \frac{Q}{m} = \frac{J}{Kg}$

2) $n_2 = n_1 \frac{U_1}{U_2}$

$n_2 = n_1 \left[\frac{M_1}{M_2}\right]^1 \left[\frac{L_1}{L_2}\right]^{-3}$

$n_2 = 0.625 \left[\frac{1\,gm}{10^3 gm}\right]^1 \left[\frac{1\,cm}{10^2 cm}\right]^{-3}$

$n_2 = 0.625 \times 10^{-3} \times 10^6$

$n_2 = 625$

3) a = 0.92 and b = 0.08

a + b = 0.92 + 0.08 = 1.00 (3 S.F.)

a – b = 0.92 – 0.08 = 0.84 (2 S.F)

4) $Q = \frac{x^2y^3}{z}$

$\frac{\Delta Q}{Q} \times 100 = 2\frac{\Delta x}{x} \times 100 + 3\frac{\Delta y}{y} \times 100 + \frac{\Delta z}{z} \times 100$

= 2(1) + 3(2) + 3

= 11%

5) V. C = LC = 0.1 mm, ZE = – 0.04 cm

During measurement :

MSR = 2.5 cm, VSR = 6th div × 0.01 mm = 0.06 cm

Corrected reading = 2.5 cm + 0.06 cm – (–0.04 cm)

= 2.60 cm.

6)

Explain Question : As given data and figure to find reading of screw gauge.

Concept : This question is based on screw gauge.

Solution :

Pitch = 2 mm $L.C = \frac{2mm}{50} = 0.04\,mm$

Linear scale = $14 \times \frac{1}{2} mm$ = 7 mm

Reading = 7 mm + 45 × 0.04 mm = 8.8 mm

Final Answer : option (3)

7) **Question Explanation:** To find the physical quantity represented by unknowns.
Concept: Dimensional analysis.
Solution: According to the principle of dimensional homogeneity.
Dimension of force = dimension of ax = dimension of bt^2.

$\Rightarrow F = ax \Rightarrow a = \frac{F}{x} = \frac{[M^1L^1T^{-2}]}{[L^1]} = [M^1L^0T^{-2}]$

and $F = bt^2 \Rightarrow b = \frac{F}{t^2} = \frac{[M^1L^1T^{-2}]}{[T^2]} = [M^1L^1T^{-4}]$

8) L.C of measurement = 0.01 cm

L.C of vernier scale = $\frac{1cm}{10 \times 10}$ = 0.01 cm

9)

Same dimensions can be added or subtracted. so options (1) and (2) are incorrect.
power of e should be dimensionless, so option (4) is incorrect.
different dimensions can be multiplied or can be divided.
so, option (3) is correct answer.

10)

$x = (t+2)^{-1} \Rightarrow x \propto (t+2)^{-1}$

$\Rightarrow v \propto (t+2)^{-2}$

$\Rightarrow a \propto (t+2)^{-3} \Rightarrow a \propto v^{3/2}$

11)

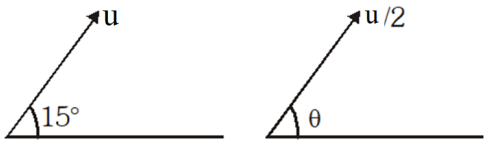
$T = t_1 + t_2$

$T = \sqrt{\frac{2h}{g}} + \frac{h}{2V}$

12) $R_1 = R_2$

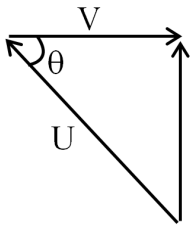
$\frac{u^2 \sin 2\theta_1}{g} = \frac{(u/2)^2 \sin 2\theta_2}{g}$

$u^2 \sin 30° = \frac{u^2}{4} \sin 2\theta_2$



$$\sin 2\theta_2 = \frac{1}{8}$$
$$2\theta = \sin^{-1}\left(\frac{1}{8}\right)$$
$$\theta = \frac{1}{2}\sin^{-1}\left(\frac{1}{8}\right)$$

13) $\frac{t_{ascent}}{t_{descent}} = \sqrt{\frac{g+a}{g-a}} = \sqrt{\frac{12}{8}} = \sqrt{\frac{3}{2}}\sqrt{\frac{\alpha}{\beta}} = \sqrt{\frac{3}{2}}$ $\alpha + \beta = 3 + 2 = 5$



14)
Velocity of block B w.r.t ground
$$\cos\theta = \frac{V}{U}$$

15) $\vec{V} = V_x\hat{i} + V_y\hat{j}$
$\vec{V} = 10\sqrt{3}i + (10 - g(1)\hat{j}$
$\vec{V} = 10\sqrt{3}i$
$\vec{r} = u_x t\hat{i} + \left(u_y t - \frac{1}{2}gt^2\right)\hat{j}$
$\vec{r} = 10\sqrt{3}(1)i + \left(10 \times 1 - \frac{10}{2}(1)^2\right)\hat{j}$
$\vec{r} = 10\sqrt{3}\hat{j} + 5\hat{j}$
$\cos\theta = \frac{\vec{V}.\vec{r}}{|V||r|} = \frac{300}{10\sqrt{3} \times \sqrt{325}} = 2\sqrt{\frac{3}{13}}$

16) In projectile motion constant acceleration is making some angle other than 0° and 180° with velocity. So both statements are true.

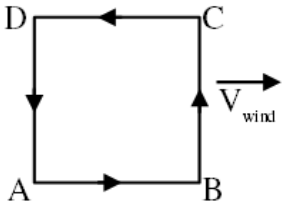
17)

→ In moving horizontal distance 5m, the ball will fall by distance $\frac{1}{2}gt^2$.
→ Time taken by particle to move 5m is,
$$t = \frac{5}{10\cos 37^\circ} = \frac{5}{8}\sec$$
→ Distance travelled by particle from C to D is,
$$y_0 = \frac{1}{2}g\left(\frac{5}{8}\right)^2 = \frac{1}{2} \times 10 \times \frac{25}{64} = \frac{125}{64} = 1.95m$$

18)

$V_1 \rightarrow$ velocity of Preeti
$V_2 \rightarrow$ velocity of escalator
$\square \rightarrow$ distance

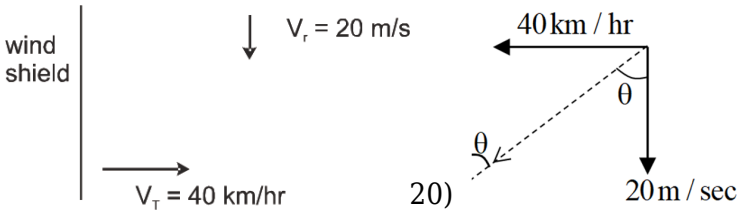
$$t = \frac{\ell}{V_1 - V_2} = \frac{\ell}{\frac{\ell}{t_1} - \frac{\ell}{t_2}} = \frac{t_1 t_2}{t_1 - t_2}$$



19) $T = t_{AB} + t_{BC} + t_{CD} + t_{DA}$

$$T = \frac{160}{50+30} + \frac{160}{\sqrt{50^2 - 30^2}} + \frac{160}{50-30} + \frac{160}{\sqrt{50^2 - 30^2}}$$

$T = 2 + 4 + 8 + 4 = 18$sec.



20)

$$\tan\theta = \frac{V_T}{V_r} = \frac{\left(40 \times \frac{5}{18}\right) \text{m/s}}{20 \text{ m/s}}$$

$$\tan\theta = \frac{5}{9}$$

$$\theta = \tan^{-1}\left(\frac{5}{9}\right)$$

21) $$\frac{2u\sin\theta}{g} = \sqrt{\frac{2H}{g}}$$

$$\sqrt{3} = \sqrt{\frac{H}{5}}$$

$$3 = \frac{H}{5}$$

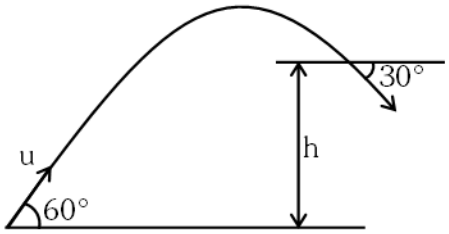
$H = 15$ m

22) $\vec{V}_{BA} = (\vec{V}_B - \vec{V}_A)$

$\vec{V}_{BA} = (10\hat{i} + 3\hat{j}) - (20\hat{i} + 4\hat{j})$

$\vec{V}_{BA} = (10\hat{i} + 3\hat{j} - 20\hat{i} + 4\hat{j})$
$\vec{V}_{BA} = -10\hat{i} - \hat{j}$

23)



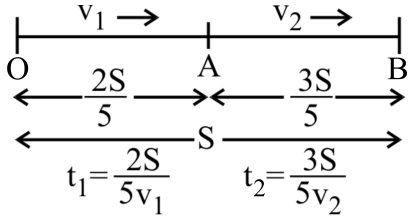
u cos 60° = v cos 30°
$v = \frac{u}{\sqrt{3}}$
In vertical direction
$v^2 = u^2 + 2as$
$\left(\frac{V}{2}\right)^2 = \left(\frac{\sqrt{3}}{2}u\right)^2 - 2gh$
$u = \sqrt{3gh}$

24)

$u_y = u\sin\theta = 25\frac{m}{sec}$
$-70 = +25t - \frac{1}{2} \times 10 \times t^2$
$5t^2 - 25t - 70 = 0$
$t^2 - 5t - 14 = 0$
$(t - 7)(t + 2) = 0$
$t = 7$ sec

25) $\vec{V}_1 = 90\hat{i} - gt\hat{j}$
$\vec{V}_2 = -40\hat{i} - gt\hat{j}$
$\vec{V}_1 . \vec{V}_2 = -90 \times 40 + g^2t^2 = 0$
$g^2t^2 = 90 \times 40$
$t^2 = 36$
$t = 6$ sec

26)



Total time = $t_1 + t_2$

$v_{avg} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{S}{\frac{2S}{5v_1} + \frac{3S}{5v_2}} = \frac{5v_1v_2S}{2Sv_2 + 3Sv_1}$

$$v_{avg} = \frac{5v_1v_2}{2v_2 + 3v_1}$$

27)

$\vec{u} = \hat{i} + 2\hat{j}$
$u_x = 1 = u\cos\theta$
$u_y = 2 = u\sin\theta$
$\tan\theta = 2$

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$
$$y = 2x - \frac{10 \times x^2}{2 \times 1}$$
$y = 2x - 5x^2$

28)

$$y = \frac{3x}{4}\left(1 - \frac{x}{240}\right)$$
$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$
$\tan\theta = 3/4$

$\theta = 37^\circ$
$R = 240$
$$R = \frac{2u^2\sin 37^\circ\cos 37^\circ}{10} = 240$$
$u = 50$
$u_x = u\cos\theta = 40$
$u_y = u\sin\theta = 30$
$$T = \frac{2U_y}{g} = 6$$
$$H_{max} = \frac{U_y^2}{2y} = 45$$

29)

Travelled distance never decreases.

30) $\frac{V_1}{V_2} = \frac{2}{3}$ $V = \sqrt{2gh}$
$V \propto \sqrt{h}$
$V^2 \propto h$
$$\frac{h_1}{h_2} = \left(\frac{V_1}{V_2}\right)^2 = \frac{4}{9}$$
time taken $t = \sqrt{\frac{2h}{g}}$
$t \propto \sqrt{h} \propto 4$

$$\frac{t_1}{t_2} = \frac{v_1}{v_2} = \frac{2}{3}$$

31) $\frac{dV}{dt} = -a\sqrt{v}$

$$\frac{dV}{v^{1/2}} = -adt$$

$$\int v^{-1/2}dv = \int -adt$$

$$2\sqrt{v} = -at + c$$

at t = 0 v = V_0

$$2\sqrt{V_0} = c$$

$$\Rightarrow 2\sqrt{v} = -at + 2\sqrt{V_0}$$

put v = 0

$$t = \frac{2\sqrt{V_0}}{a}$$

32)

When partical crosses y axis x = 0, $x_i = 0$, $x_f = 0 \Rightarrow \Delta S_x = 0$

$$S_x = 11t - \frac{1}{2} \times \frac{22}{5} \times t^2 = 0$$

t = 5 sec

$$\vec{V}_5 = (11\hat{i} + 14\hat{j}) + \left(\frac{-22}{5}\hat{i} + \frac{2}{15}\hat{j}\right) \times 5$$

$$\vec{V}_5 = -11\hat{i} + \frac{44}{3}\hat{j}$$

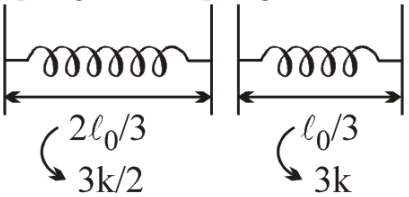
$$\tan\theta_x = \frac{44}{3 \times 11} = \frac{4}{3}$$

θ = 53° from –x axis towards +y axis.

33) $\text{Distance} \geq |\text{Displacement}|$

34)

When a spring is cut into two parts each part has spring constant more than that of original spring. If k = spring constant & ℓ_0 = natural length, then for cut parts



If they are stretched by same amount then work done in shorter part will be double than that in the case of longer part.

Because, $W = \frac{1}{2}kx^2$

35) KE ∝ P^2 Let P_i = P

$P_f = P + \frac{25P}{100} = \frac{5}{6}P$

$\frac{KE_f - KE_i}{KE_i} \times 100\% = \frac{P_f^2 - P_i^2}{P_i^2} \times 100\%$

= 56%

36) W.D. = Area under F-x curve

$\frac{1}{2}(4)(10) + \frac{1}{2}(4)(10) + 40 = \Delta K$

$80 = \frac{1}{2}(0.1)V^2$

$\boxed{V = 40\,m/s}$

37) F.dx = dK

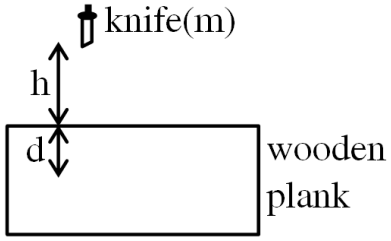
$\frac{dK}{dx} = F$

Shope of K-x cursive gives force.

So slope is max at D, hence acc. is max. at D.

38) May be positive or Negative

39) Let F be the resistance force of wooden block



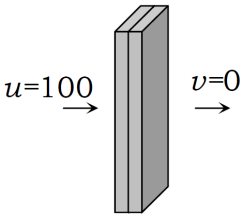
$W_{mg} + W_F = KE_f - KF_i$

mg (h + d) + F (d) (–1) = 0 – 0

mg (h + d) = Fd

$F = mg\left(1 + \frac{h}{d}\right)$

40)



$F \times 2S = \frac{1}{2}mu^2$

$F \times nS = \frac{1}{2}m(2u)^2$

$\frac{2}{n} = \frac{1}{4} \Rightarrow n = 8$

41)

For conservative force.
- Work done is independent on path followed.
- Work done in closed path is zero
- depend on initial and final position
- $F = -\frac{dv}{dr}$, work done against conservative force is stored in the form of potential energy.

42) $\vec{F} = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]$

$\vec{F} = -\left[\frac{20y}{z}\hat{i} + \frac{20x}{z}\hat{j} - \frac{20xy}{z^2}\hat{k}\right]$

$\vec{F} = -\left(\frac{20y}{z}\right)\hat{i} - \left(\frac{20x}{z}\right)\hat{j} + \left(\frac{20xy}{z^2}\right)\hat{k}$

43) $\cos 60^\circ = \frac{L}{L+x}$

$L + x = 2L$

$X = L$

$\frac{1}{2}kL^2 = \frac{1}{2}mv^2$

$v = L\sqrt{\frac{k}{m}}$

44) $F \propto x^{-1/3}$

$\Rightarrow a \propto x^{-1/3}$

$\Rightarrow \int vdv \propto \int x^{-1/3}dx$

$\Rightarrow v^2 \propto x^{2/3}$

$\Rightarrow v \propto x^{1/3}$

power = F.v

$p \propto x^{-1/3}.x^{1/3}$

$\Rightarrow$ power = const. $\propto x^0$

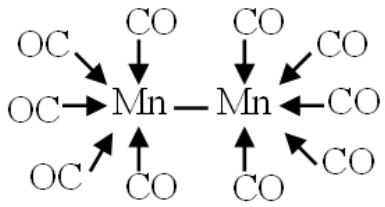
45) $p_1 = \frac{m_1 gh}{t_1}$

$p_2 = \frac{m_2 gh}{t_2}$

$\frac{p_1}{p_2} = \frac{m_1}{m_2} \cdot \frac{t_2}{t_1} = \frac{4}{3} \cdot \frac{11}{12}$

$= \frac{11}{9}$

CHEMISTRY



46)
NCERT Pg. # 136

47)

NEW NCERT XII, Part-I / Pg # 136 - 137

48) Non-electrolytes that's why it show minimum conductivity among following options.

49)

Strength of ligand ↑ Δ_0 ↑ ($\Delta_t < \Delta_0$)
Absorbed Energy ↑ $\lambda_{absorbed}$ ↓

50)

(A) $[NiCl_4]^{2-}$ ⇒ Oxidation state = +2
Electronic configuration = $3d^8\ 4s^0$
Cl = W.F.L. No Pairing

⇅	⇅	⇅	1	1

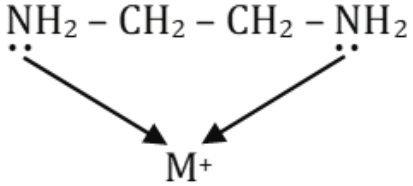
2 unpaired electrons
(D) $[Ni(H_2O)_6]^{2+}$
Oxidation state = +2, configuration = $3d^8$
H_2O is W.F.L. No pairing

⇅	⇅	⇅	1	1

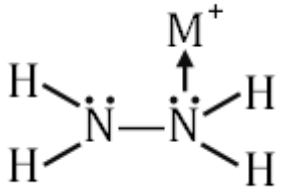
2 unpaired electrons
Hence both are paramagnetic

51)

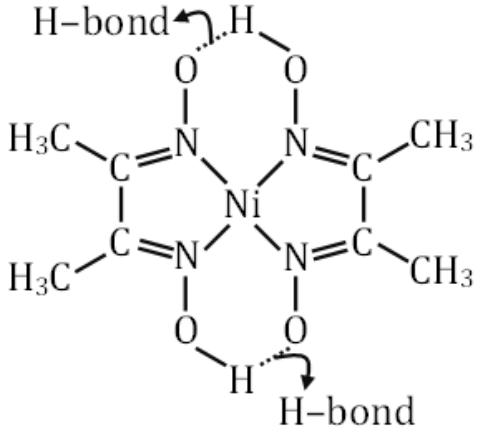
Explaining : Classify the ligands
Concept : This question is based on Classification of ligands
Solution :
(A) Ethylenediamine with two donor site (bidentate)



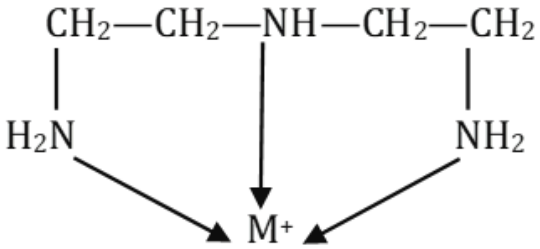
(B) Hydrazine with only one donor site or site (Monodentate ligand)



(C) Dimethylgyloximato (dmg^-) can form hydrogen bond

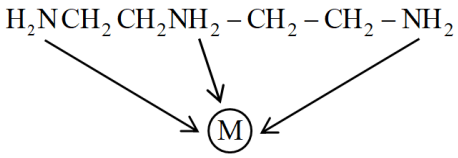


(D) Dien with three donor site (Tridentate ligand)



Final Answer : (3)

52) New NCERT-12th Part-1, Pg # 133, Chapter - Coordination compound



53)
One dien forms two rings
So, rings = 4, O.S. of Cu = +2
So, 4 + 2 = 6

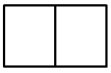
54)

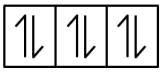
CFSE = $\left(-\frac{2}{5}x+\frac{3}{5}y\right)\Delta_0$

$\left[d^4=t_{2g}^3eg^1,\ x=3, y=1\right]$

$=\left(-\frac{2}{5}\times 3+\frac{3}{5}\times 1\right)\Delta_0 = -\frac{3}{5}\Delta_0 = -0.6\Delta_0$

55) $C_2O_4^{-2}$ behave as SFL with Co^{+3}



Co^{+3}, d^6  unpaired $e^- = 0$

56)

NEW NCERT 12th Part I / Pg # (123 – 124)

57) New NCERT-XII, Part I / Pg # 137

58)

Ma_4b_2 and Ma_3b_3 shows G.I.

59)

vacant π* M.O. of ligands are used in synergic bonding.

60) $\Delta_{SE} = \frac{hc}{\lambda}$

$\Delta_{SE} \propto \frac{1}{\lambda} \propto \oplus$ve charge on CMA $\propto \frac{1}{\text{Strength of ligand}}$

61) $\left.\begin{matrix} Ma_2b_2 \text{ (sq. PL)} \\ Ma_4b_2 \text{ (O.H.)} \end{matrix}\right]$ Shows stereomerism

Ma_5b (O.H.) Not shows

62)

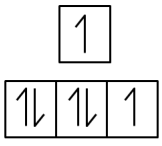
$K[PtCl_3(C_2H_4)]$
π e^- s of C_2H_4 attach with Pt
O.S. of Pt = +2

63)

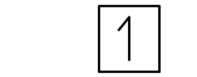
Stability constant ∝ Stability
Stability ∝ Strength of ligand

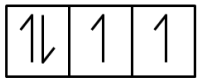
64)

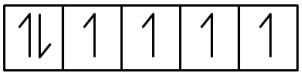
If both ions are complex, then shows coordination
(1) Isomerism
(2) $\Delta_{sp} > \Delta_0 > \Delta_t$

(3)  1 unpaired e^-

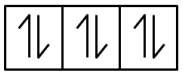
65)

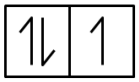
 unpaired e^-

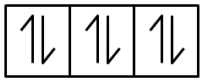
(1) d^4  = 1

(2)  = 4



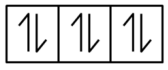
(3) d^6  = 0



(4) d^9  = 1

66)



Co^{+3} d^6  unpaired e^- = 0

67)

New NCERT 12th part I / Pg # 131-132

68)

New NCERT 12th part I / Pg # 131

69) New NCERT 12th part I / Pg # 133

70)

One or two d vacant d orbitals required for inner orbital complex

$t_{2g}^3 e_g^0$, $t_{2g}^6 e_g^1$

71)

$[M(aa)_2b_2]$ shows O.I. due to NO_2 shows linkage also shows G.I. and ionisation
$(Co(NO_2)_2(en)_2)Br$

72)

New NCERT 12th part I / Pg # 132

73)

$(Fe(NH_3)_x(CN)_y)^{-1}$
conf. t_{2g}^5, eg^0 (given)
So Fe^{+3},
$+3 + x(0) + y(-1) = -1 \Rightarrow y = 4$
C.N = 6 so, x = 2, y = 4 ⇒ x + y = 6

74)

$Na_2[Fe(CN)_5(NO)]$
(Fe^{+2}, NO^+) C.N. = 6
EAN = 26 – 2 + 2 (6) = 36

75)

$(Cr(CO)_6)$
EAN = 24 – 0 + 2(6) = 36
is At. No of Noble gas
So, it is stable

76) P_V = O.S. of CMA
S_V - C.N. of complex

77) Due to synergic bonding ⊖ve charge ↑
⇒ synergic bonding ↑
M – C B.O ↑ ⇒ C – O B.O ↓
B.O. $CO^+ > CO > Fe(CO)_5 > Mn(CO)_5^-$
$H_3B^- \leftarrow CO^+$
So, CO^+

78)

New NCERT XIIth / Part I / Pg # 123, IUPAC name

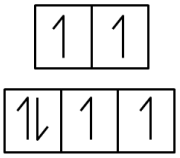
79)

$[M(aa.)_2b_2]$ and $[M(aa)_2bc]$ two G.I. and cis is optically active so total stereo Iso = 3

80)

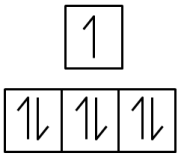
$\mu = 4.9 \Rightarrow$ unpaired $e^- = 4$

Fe^{+2} d^6



$\mu = 0 \Rightarrow$ unpaired $e^- = 0$

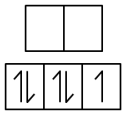
d^6



81) due to $\Delta_t <$ P.E.

82)

Fe^{+3}, d^5



S.L. d^2sp^3, Para, I.O.C.

83)

$$dmg + Ni^{2} \xrightarrow[(pH>7)]{\text{Basic Medium}} \text{Red ppt}$$

84)

Rings increases, So stability ↑

85)

Stability ∝ No. of chelate rings
∝ Strength of ligand
∝ +ve charge on metal
3d < 4d < 5d

86)

New NCERT XII, Part - I / Pg # 119 - 120

87)

NCERT XII, Part - I / Pg # 121

88)

Ni^{+2}, C.N. = 4, S.F.L., dsp^2, sq. pL
More ⊖ ve charge

89)

More ⊖ve charge on metal
M – C B.O. ↑ more ⇒ B.O. ↓
C – O B.L. ↑ more
So, $[Fe(CO)_4]^{2-}$ C – O, B.L. is more.

90)

New NCERT XII, Part - I / Pg # 136